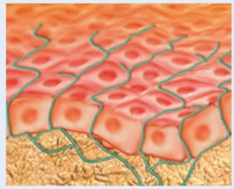
SKIN MICRORECEPTORS

Deformation of the layers within a receptor, and expansion or contraction as a result of temperature changes, generate nerve impulses. The impulses travel along the receptor's nerve fibre, which joins with bundles of other fibres in the deep dermis or tissue layers below. Most receptors "fire" nerve signals infrequently and irregularly when not stimulated, increasing their firing rate when the skin is touched.

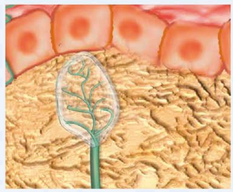
TYPES OF SENSOR

Each type of microsensor is set at a particular depth in the dermis that best suits its function. The largest receptors, Pacinian corpuscles, are located at the deepest level, near the base of the dermis. Sensors for light touch are located near or just within the epidermal layer.

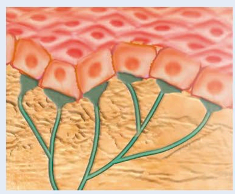
Free nerve endings
Branching, usually unsheathed sensors of temperature, light touch, pressure, and pain. They are found all over the body and in all types of connective tissue.



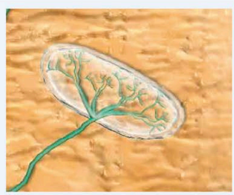
Meissner's corpuscle
Encapsulated nerve ending in the skin's upper dermis, especially on the palms, soles, lips, eyelids, external genitals, and nipples. It responds to light pressure.



Merkel's disc
Naked (unencapsulated) receptors, usually in the upper dermis or lower epidermis, especially in non-hairy areas. They sense faint touch and light pressure.



Ruffini corpuscle
Encapsulated receptor in the skin and deeper tissue that reacts to continuous touch and pressure. In joint capsules, it responds to rotational movement.



Pacinian corpuscle
Large, covered receptor located deep in the dermis, as well as in the bladder wall, and near joints and muscles. It senses stronger, more sustained pressure.